$$\begin{array}{c|cc}
 & \mathcal{K}_{0^+}^{\#1} & \mathcal{K}_{0^-}^{\#1\,\alpha\beta\chi} \\
\hline
\mathcal{K}_{0^+}^{\#1\,\dagger} & 0 & 0 \\
\mathcal{K}_{0^-}^{\#1\,\dagger\alpha\beta\chi} & 0 & 0
\end{array}$$

$$\begin{array}{c|cccc}
 & \mathcal{K}_{1^+}^{\#1\,\alpha\beta} & \mathcal{K}_{1^+}^{\#2\,\alpha\beta} & \mathcal{K}_{1^-}^{\#1\,\alpha} & \mathcal{K}_{1^-}^{\#2\,\alpha} \\
\hline
\mathcal{K}_{1^+}^{\#1\,\dagger\alpha\beta} & -\frac{\Upsilon_1 k^2}{2} & -\frac{\Upsilon_2 k^2}{2\sqrt{2}} & 0 & 0 \\
\mathcal{K}_{1^+}^{\#2\,\dagger\alpha\beta} & -\frac{\Upsilon_2 k^2}{2\sqrt{2}} & -\frac{\Upsilon_3 k^2}{4} & 0 & 0 \\
\mathcal{K}_{1^-}^{\#1\,\dagger\alpha} & 0 & 0 & \frac{\Upsilon_4 k^2}{2} & -\frac{\Upsilon_4 k^2}{2\sqrt{2}} \\
\mathcal{K}_{1^-}^{\#2\,\dagger\alpha} & 0 & 0 & -\frac{\Upsilon_4 k^2}{2\sqrt{2}} & \frac{\Upsilon_4 k^2}{4}
\end{array}$$

$$\begin{array}{c|cc}
 & \mathcal{K}_{2^+}^{\#1\,\alpha\beta} & \mathcal{K}_{2^-}^{\#1\,\alpha\beta\chi} \\
\hline
\mathcal{K}_{2^+}^{\#1\,\dagger\alpha\beta} & \frac{3\Upsilon_5 k^2}{2} & 0 \\
\mathcal{K}_{2^-}^{\#1\,\dagger\alpha\beta\chi} & 0 & \frac{3k^2\,{}^{(4)}\kappa_{15}}{2}
\end{array}$$

$$\begin{array}{c|cc}
 & \mathcal{J}_{0^+}^{\#1} & \mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi} \\
\hline
\mathcal{J}_{0^+}^{\#1\,\dagger} & 0 & 0 \\
\mathcal{J}_{0^-}^{\#1\,\dagger\alpha\beta\chi} & 0 & 0
\end{array}$$

$$\begin{array}{c|cccc}
 & \mathcal{J}_{1^+}^{\#1\,\alpha\beta} & \mathcal{J}_{1^+}^{\#2\,\alpha\beta} & \mathcal{J}_{1^-}^{\#1\,\alpha} & \mathcal{J}_{1^-}^{\#2\,\alpha} \\
\hline
\mathcal{J}_{1^+}^{\#1\,\dagger\alpha\beta} & -\frac{\Upsilon_3 k^2}{4\,\mathrm{Det}(1^+)} & \frac{\Upsilon_2 k^2}{2\sqrt{2}\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}_{1^+}^{\#2\,\dagger\alpha\beta} & \frac{\Upsilon_2 k^2}{2\sqrt{2}\,\mathrm{Det}(1^+)} & -\frac{\Upsilon_1 k^2}{2\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}_{1^-}^{\#1\,\dagger\alpha} & 0 & 0 & \frac{2}{3\,\mathrm{Det}(1^-)} & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} \\
\mathcal{J}_{1^-}^{\#2\,\dagger\alpha} & 0 & 0 & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} & \frac{1}{3\,\mathrm{Det}(1^-)}
\end{array}$$

$$\begin{array}{c|cc}
 & \mathcal{J}_{2^+}^{\#1\,\alpha\beta} & \mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi} \\
\hline
\mathcal{J}_{2^+}^{\#1\,\dagger\alpha\beta} & \frac{1}{\mathrm{Det}(2^+)} & 0 \\
\mathcal{J}_{2^-}^{\#1\,\dagger\alpha\beta\chi} & 0 & \frac{2}{3k^2\,{}^{(4)}\kappa_{15}}
\end{array}$$

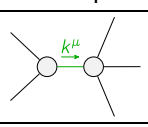
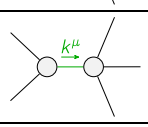
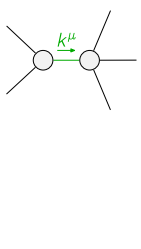
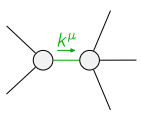
Abbreviations used in matrices

$$\Upsilon_1 == 3\,{}^{(4)}\kappa_1 - 4\,{}^{(4)}\kappa_{15} + 3\,{}^{(4)}\kappa_2 + 2\,{}^{(4)}\kappa_3 \;\&\&\; \Upsilon_2 == 2\,{}^{(4)}\kappa_{15} - {}^{(4)}\kappa_4 \;\&\&$$
$$\Upsilon_3 == 5\,{}^{(4)}\kappa_{15} - 2\,({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) \;\&\&\; \Upsilon_4 == 3\,{}^{(4)}\kappa_{15} - 2\,{}^{(4)}\kappa_2 \;\&\&\; \Upsilon_5 == {}^{(4)}\kappa_1 + {}^{(4)}\kappa_2 \;\&\&$$
$$\mathrm{Det}(1^+) == \frac{1}{8}k^4(15\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15} - 24\,{}^{(4)}\kappa_{15}^{\,2} - 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 - 4\,{}^{(4)}\kappa_3^{\,2} - 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_4 - 4\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 -$$
$${}^{(4)}\kappa_4^{\,2} - 6\,{}^{(4)}\kappa_1({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + 3\,{}^{(4)}\kappa_{15}(5\,{}^{(4)}\kappa_2 + 6\,{}^{(4)}\kappa_3 + 4\,{}^{(4)}\kappa_4)) \;\&\&\; \mathrm{Det}(2^+) == \frac{3}{2}k^2({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)$$

Lagrangian

$$\begin{aligned}
&{}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + \\
&{}^{(4)}\kappa_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - 3\,{}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} + 3\,{}^{(4)}\kappa_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - 3\,{}^{(4)}\kappa_2\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \\
&{}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - 3\,{}^{(4)}\kappa_{15}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \frac{9}{4}\,{}^{(4)}\kappa_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \\
&\frac{1}{2}\,{}^{(4)}\kappa_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\,{}^{(4)}\kappa_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + {}^{(4)}\kappa_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} + {}^{(4)}\kappa_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\,\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}_{1^-}^{\#1\,\alpha} + 2\,\mathcal{J}_{1^-}^{\#2\,\alpha} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
Total # constraint(s):	5	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{{}^{(4)}\kappa_1 + {}^{(4)}\kappa_{15} + {}^{(4)}\kappa_2}{{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2}$
	1	0	$-\frac{{}^{(4)}\kappa_1 + {}^{(4)}\kappa_{15} + {}^{(4)}\kappa_2}{{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2}$
	1	0	$-((-72\,{}^{(4)}\kappa_1^{\,2}\,{}^{(4)}\kappa_{15}^{\,2} + 72\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}^{\,3} + 48\,{}^{(4)}\kappa_1^{\,2}\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2 - 192\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_2 + 72\,{}^{(4)}\kappa_{15}^{\,3}\,{}^{(4)}\kappa_2 + 96\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2^{\,2} -$ $120\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_2^{\,2} + 48\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2^{\,3} - 48\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_3 + 32\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 - 48\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 +$ $32\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2^{\,2}\,{}^{(4)}\kappa_3 + 12\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_4 - 8\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_4 + 12\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_4 - 8\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2^{\,2}\,{}^{(4)}\kappa_4 +$ $\sqrt{((3\,{}^{(4)}\kappa_{15} - 2\,{}^{(4)}\kappa_2)^2({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)^2(3177\,{}^{(4)}\kappa_{15}^{\,4} - 6\,{}^{(4)}\kappa_{15}^{\,3}(789\,{}^{(4)}\kappa_2 + 676\,{}^{(4)}\kappa_3 + 146\,{}^{(4)}\kappa_4) +$ $9\,{}^{(4)}\kappa_1^{\,2}(209\,{}^{(4)}\kappa_{15}^{\,2} - 36\,{}^{(4)}\kappa_{15}({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + 4({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2) + (6\,{}^{(4)}\kappa_2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + (2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2)^2 +$ $3\,{}^{(4)}\kappa_{15}^{\,2}(627\,{}^{(4)}\kappa_2^{\,2} + 552\,{}^{(4)}\kappa_3^{\,2} + 296\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + 106\,{}^{(4)}\kappa_4^{\,2} + 4\,{}^{(4)}\kappa_2(299\,{}^{(4)}\kappa_3 + 76\,{}^{(4)}\kappa_4)) -$ $2\,{}^{(4)}\kappa_{15}(162\,{}^{(4)}\kappa_2^{\,2}({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + 14(2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^3 + 3\,{}^{(4)}\kappa_2(92\,{}^{(4)}\kappa_3^{\,2} + 120\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + 37\,{}^{(4)}\kappa_4^{\,2})) -$ $6\,{}^{(4)}\kappa_1(789\,{}^{(4)}\kappa_{15}^{\,3} - {}^{(4)}\kappa_{15}^{\,2}(627\,{}^{(4)}\kappa_2 + 598\,{}^{(4)}\kappa_3 + 152\,{}^{(4)}\kappa_4) + {}^{(4)}\kappa_{15}(92\,{}^{(4)}\kappa_3^{\,2} + 120\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 +$ $37\,{}^{(4)}\kappa_4^{\,2} + 108\,{}^{(4)}\kappa_2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)) - 2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)(6\,{}^{(4)}\kappa_2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + (2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2)))))/$ $({}^{(4)}\kappa_{15}(3\,{}^{(4)}\kappa_{15} - 2\,{}^{(4)}\kappa_2)({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)(-15\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15} + 24\,{}^{(4)}\kappa_{15}^{\,2} + 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + 4\,{}^{(4)}\kappa_3^{\,2} + 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_4 +$ $4\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^{\,2} + 6\,{}^{(4)}\kappa_1({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) - 3\,{}^{(4)}\kappa_{15}(5\,{}^{(4)}\kappa_2 + 6\,{}^{(4)}\kappa_3 + 4\,{}^{(4)}\kappa_4))))$
	1	0	$(72\,{}^{(4)}\kappa_1^{\,2}\,{}^{(4)}\kappa_{15}^{\,2} - 72\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}^{\,3} - 48\,{}^{(4)}\kappa_1^{\,2}\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2 + 192\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_2 - 72\,{}^{(4)}\kappa_{15}^{\,3}\,{}^{(4)}\kappa_2 - 96\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2^{\,2} +$ $120\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_2^{\,2} - 48\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2^{\,3} + 48\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_3 - 32\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + 48\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 -$ $32\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2^{\,2}\,{}^{(4)}\kappa_3 - 12\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_4 + 8\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_4 - 12\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_4 + 8\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2^{\,2}\,{}^{(4)}\kappa_4 +$ $\sqrt{((3\,{}^{(4)}\kappa_{15} - 2\,{}^{(4)}\kappa_2)^2({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)^2(3177\,{}^{(4)}\kappa_{15}^{\,4} - 6\,{}^{(4)}\kappa_{15}^{\,3}(789\,{}^{(4)}\kappa_2 + 676\,{}^{(4)}\kappa_3 + 146\,{}^{(4)}\kappa_4) +$ $9\,{}^{(4)}\kappa_1^{\,2}(209\,{}^{(4)}\kappa_{15}^{\,2} - 36\,{}^{(4)}\kappa_{15}({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + 4({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2) + (6\,{}^{(4)}\kappa_2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + (2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2)^2 +$ $3\,{}^{(4)}\kappa_{15}^{\,2}(627\,{}^{(4)}\kappa_2^{\,2} + 552\,{}^{(4)}\kappa_3^{\,2} + 296\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + 106\,{}^{(4)}\kappa_4^{\,2} + 4\,{}^{(4)}\kappa_2(299\,{}^{(4)}\kappa_3 + 76\,{}^{(4)}\kappa_4)) -$ $2\,{}^{(4)}\kappa_{15}(162\,{}^{(4)}\kappa_2^{\,2}({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + 14(2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^3 + 3\,{}^{(4)}\kappa_2(92\,{}^{(4)}\kappa_3^{\,2} + 120\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + 37\,{}^{(4)}\kappa_4^{\,2})) -$ $6\,{}^{(4)}\kappa_1(789\,{}^{(4)}\kappa_{15}^{\,3} - {}^{(4)}\kappa_{15}^{\,2}(627\,{}^{(4)}\kappa_2 + 598\,{}^{(4)}\kappa_3 + 152\,{}^{(4)}\kappa_4) + {}^{(4)}\kappa_{15}(92\,{}^{(4)}\kappa_3^{\,2} + 120\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 +$ $37\,{}^{(4)}\kappa_4^{\,2} + 108\,{}^{(4)}\kappa_2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)) - 2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)(6\,{}^{(4)}\kappa_2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + (2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2)))))/$ $({}^{(4)}\kappa_{15}(3\,{}^{(4)}\kappa_{15} - 2\,{}^{(4)}\kappa_2)({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)(-15\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15} + 24\,{}^{(4)}\kappa_{15}^{\,2} + 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + 4\,{}^{(4)}\kappa_3^{\,2} + 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_4 +$ $4\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^{\,2} + 6\,{}^{(4)}\kappa_1({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) - 3\,{}^{(4)}\kappa_{15}(5\,{}^{(4)}\kappa_2 + 6\,{}^{(4)}\kappa_3 + 4\,{}^{(4)}\kappa_4))$
	1	0	$(72\,{}^{(4)}\kappa_{15}^{\,4} + 51\,{}^{(4)}\kappa_{15}^{\,3}\,{}^{(4)}\kappa_2 - 156\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_2^{\,2} + 60\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2^{\,3} - 54\,{}^{(4)}\kappa_{15}^{\,3}\,{}^{(4)}\kappa_3 - 54\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + 96\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2^{\,2}\,{}^{(4)}\kappa_3 -$ $24\,{}^{(4)}\kappa_2^{\,3}\,{}^{(4)}\kappa_3 + 12\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_3^{\,2} + 16\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3^{\,2} - 16\,{}^{(4)}\kappa_2^{\,2}\,{}^{(4)}\kappa_3^{\,2} - 36\,{}^{(4)}\kappa_{15}^{\,3}\,{}^{(4)}\kappa_4 - 30\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_4 +$ $72\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2^{\,2}\,{}^{(4)}\kappa_4 - 24\,{}^{(4)}\kappa_2^{\,3}\,{}^{(4)}\kappa_4 + 12\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + 16\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 - 16\,{}^{(4)}\kappa_2^{\,2}\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 +$ $3\,{}^{(4)}\kappa_{15}^{\,2}\,{}^{(4)}\kappa_4^{\,2} + 4\,{}^{(4)}\kappa_{15}\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_4^{\,2} - 4\,{}^{(4)}\kappa_2^{\,2}\,{}^{(4)}\kappa_4^{\,2} - 6\,{}^{(4)}\kappa_1^{\,2}(3\,{}^{(4)}\kappa_{15} - 2\,{}^{(4)}\kappa_2)(5\,{}^{(4)}\kappa_{15} - 2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)) +$ ${}^{(4)}\kappa_1(3\,{}^{(4)}\kappa_{15} - 2\,{}^{(4)}\kappa_2)(33\,{}^{(4)}\kappa_{15}^{\,2} + 24\,{}^{(4)}\kappa_2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + 2(2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2 - 6\,{}^{(4)}\kappa_{15}(10\,{}^{(4)}\kappa_2 + 5\,{}^{(4)}\kappa_3 + 3\,{}^{(4)}\kappa_4)) -$ $\sqrt{3}\sqrt{((3\,{}^{(4)}\kappa_{15} - 2\,{}^{(4)}\kappa_2)^2(7\,{}^{(4)}\kappa_1^{\,2} + 5\,{}^{(4)}\kappa_{15}^{\,2} + 14\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_2 + 7\,{}^{(4)}\kappa_2^{\,2})(-15\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15} + 24\,{}^{(4)}\kappa_{15}^{\,2} + 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 +$ $4\,{}^{(4)}\kappa_3^{\,2} + 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_4 + 4\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^{\,2} + 6\,{}^{(4)}\kappa_1({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) - 3\,{}^{(4)}\kappa_{15}(5\,{}^{(4)}\kappa_2 + 6\,{}^{(4)}\kappa_3 + 4\,{}^{(4)}\kappa_4))^2))/$ $({}^{(4)}\kappa_{15}(3\,{}^{(4)}\kappa_{15} - 2\,{}^{(4)}\kappa_2)({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)(-15\,{}^{(4)}\kappa_1\,{}^{(4)}\kappa_{15} + 24\,{}^{(4)}\kappa_{15}^{\,2} + 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + 4\,{}^{(4)}\kappa_3^{\,2} + 6\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_4 +$ $4\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^{\,2} + 6\,{}^{(4)}\kappa_1({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) - 3\,{}^{(4)}\kappa_{15}(5\,{}^{(4)}\kappa_2 + 6\,{}^{(4)}\kappa_3 + 4\,{}^{(4)}\kappa_4)))$
	2	0	(Hidden for brevity)
	2	0	(Hidden for brevity)
	2	0	(Hidden for brevity)